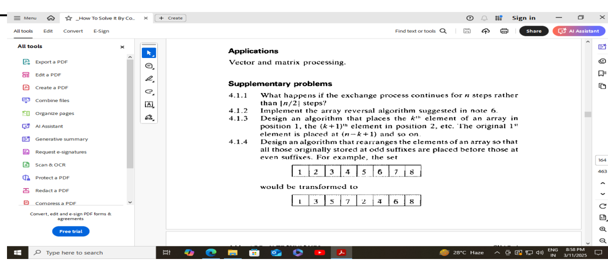
	CP-II
1	What is Data structures? What are the basic operations of data structure?
2	Write down the basic terminologies used in data structure.
3	Define Abstract Data Type and explain its fundamental concepts.
4	Write a program to create array as an abstract data type with initialization function.
5	Explain the classification of data structures and compare characteristics of linear and nonlinear data structure with example.
6	Consider a zero-indexed array A with dimensions 5 x 6 x 7 that stores 4-byte integers is located at an address 6400. Calculate the address of A (4, 5, 6) using lexicographic order (row-major order) and column major order.
7	Write an abstract function to check whether the array is full or not.
8	Write an abstract function to create an array in a single function call.
9	Write a program using a function for deleting an element in the array at specified index position.
10	Write a program using a function for inserting an element in the array at specified index position.
11	Write an array reversal algorithm using in place shifting operation.
12	What happens if the exchange process continues for n steps rather than |n/2| steps in array reversal algorithm?
13	Write an algorithm that places k th element of an array in position 1, the (k+1)th element in position 2, etc. The original 1st element is placed at (n-k+1) and so on.
14	Write an algorithm that rearranges the elements of an array so that all those originally stored at odd suffixes are placed before those at even suffixes. For example, the set



15	Consider a multi-dimensional Array $A[90][30][40]$ with base address starting at 1000. Calculate the address of A[10] [20] [30] in row major order and column major order. Assume the first element is stored at $A[2][2][2]$ and each element take 2 byte.
16	Consider a set of 100 student's examination marks (in the range 0 to 100) are given. Write a pseudo code to count the number of students who achieved each possible mark in the counter array.

17	Modify the algorithm above so that a histogram is obtained only for each ten percentile range (e.g. 0--> 10%, 11-->20%, …) rather than for each individual mark.
18	Write an algorithm to find the number of times the maximum occurs in an array of n elements. Only one pass through the array should be made.
19	Write a program using function to find the maximum element in an array along with its position.
20	Write a program using function to find the minimum element in an array along with its position.
21	Write a program using function to find the second largest element in an array of n elements.
22	Write an algorithm to find the minimum, the maximum, and how many times they both occur in an array of n elements.
23	Remove all duplicates from an ordered array and contract the array accordingly.
24	Delete from an ordered array all elements that occurs more than k times.
25	Given a randomly ordered array of n elements, partition the elements into two subsets such that elements less than or equal to x are in one subset and elements > x are in other subset.
26	**Consider the linear arrays AAA [5 : 50], BBB [– 5 : 10] and CCC [1 : 8].** a) Find the number of elements in each array. b) Suppose base (AAA) = 300 and w = 4 words per memory cell for AAA. c) Find the address of AAA [15], AAA [35] and AAA [55].
27	Suppose multidimensional arrays P and Q are declared as P(– 2: 2, 2: 22) and Q(1: 8, – 5: 5, – 10 : 5) stored in column major order
	i. Find the length of each dimension of P and Q. ii. The number of elements in P and Q. iii. Assuming base address (Q) = 400, W = 4, find the effective indices E1 , E2 , E3 and address of the element Q[3, 3, 3].